

Name: _____

Class: _____

Date: _____

Book Worksheet 3

Machines That Calculate



Read: 1600s–1930s: early calculators, punched cards, and the first “programs”

The History of the Computer · Rachel Ignatofsky · Pages _____ to _____



The Big Picture

Inventors built machines with gears and punched cards to calculate and follow instructions. Their ideas — memory, a processor, and programs — are still inside every computer.

Part 1 • Influential People

Fill in the table as you read.

Who / What	What did they create or do?	Why did it matter?
Blaise Pascal		
Gottfried Leibniz		
Joseph Marie Jacquard		
Charles Babbage		
Ada Lovelace		
Herman Hollerith		

Part 2 • Fill It In

Use each word from the word bank once.

binary

1890

CPU

punched cards

IBM

Ada Lovelace

- Jacquard's loom used _____ to control the pattern being woven.
- Babbage's Analytical Engine had a “mill,” like today's _____, and a “store,” like today's memory.
- _____ is often called the first computer programmer.
- Hollerith's machines helped count the _____ U.S. Census much faster.
- Hollerith's company eventually became part of _____.
- Leibniz wrote about the _____ number system, which uses only 0 and 1.

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Check Your Understanding

Part 3 • Timeline

Order these events from earliest (1) to latest (7). Write the number in the box.

- | | |
|--|--|
| ☐ A. Pascal builds the Pascaline | ☐ B. Babbage designs the Analytical Engine |
| ☐ C. Leibniz writes about binary numbers | ☐ D. Alan Turing describes the “Turing machine” |
| ☐ E. Ada Lovelace publishes her notes and algorithm | ☐ F. Jacquard’s punched-card loom |
| ☐ G. Hollerith’s machines count the U.S. Census | |

Part 4 • Check Your Understanding

1. Which invention is most like a computer program?

- A. The gears inside the Pascaline
- B. Tally marks on a notched bone
- C. The beads on a soroban
- D. Punched cards that told the Jacquard loom what pattern to weave

2. Why is Ada Lovelace remembered as a computing pioneer?

- A. She built the first electronic computer
- B. She wrote an algorithm for the Analytical Engine and saw it could work with more than numbers
- C. She founded IBM
- D. She invented the abacus

3. Hollerith’s punched cards are an early example of...

- A. Storing data so a machine can read and process it
- B. Artificial intelligence
- C. Wireless networking
- D. Encryption

Part 5 • Stories from History

Pick one story or person from this section that surprised you. **Who** was involved, **what** happened, and **why** does it still matter?

Part 6 • Unit 5.1 Connection

When Jacquard’s loom spread, some weavers feared losing their jobs and even attacked the machines. Compare this to automation and AI today. Describe one **beneficial** and one **harmful** effect of machines that do work people used to do.
